

Comments on the reviewing report for Invariant measures for the stochastic heat equation with no drift term

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We thank the anonymous referee for careful reading of the paper, some helpful comments, and especially pointing out the reference by Gu and Li [1]. In this revision, we have carefully taken into all comments from the report. The list of major changes are as follows:

1. pages 3 – 4, in the both paragraphs after Eq. (1.9), we have included the reference [1] and made some comparison with our current work. As a short summary, the focus of our paper is on the conditions on the correlation function/measure and initial conditions in order to guarantee the existence of invariant measures. The focus of the paper [1] is different, and convenient conditions were made in [1] for the correlation function, which is assumed to be smooth function with compact support. The initial condition in [1] is essentially constant initial data with some perturbation. In contrast, our conditions on both correlation function/measure and initial conditions are much weaker; see the main theorem – Theorem 1.5 of the paper.
2. page 5, Figure 1 has been simplified and updated. We postponed the conditions formulated using $\mathcal{H}_\alpha(t)$ back in Section 3.1 when it is first used.
3. page 5, Remark 1.2 has been added to comment on the including the *strengthened Dalang's condition* in our main theorem.
4. page 7, Remark 1.6 has been included to comment on our conditions on the noise. Both this change and the previous one are related to one comment from the referee about our additional assumption, namely, the *strengthened Dalang's condition*.
5. Section 3.2, Figure 2 and Remark 3.5 have been created to help the reader. We have also updated Lemma 3.6 accordingly to reflect the changes in items 2, 3, and 4 above.

6. In the Acknowledgment section, we include acknowledgment to Yu Gu for the discussions on the reference [\[1\]](#).

References

- [1] Yu Gu and Jiawei Li. “Fluctuations of a nonlinear stochastic heat equation in dimensions three and higher”. In: *SIAM J. Math. Anal.* 52.6 (2020), pp. 5422–5440. issn: 0036-1410.

Below is the reviewing report from the anonymous referee:

The authors consider the stochastic heat equation (SHE) on R^d where the driving noise is white in time and colored in space. The main result of this paper is to provide some conditions on the initial data and the correlation measure (or the spectral measure) of the noise that guarantees the existence of an invariant measure in some weighted L^2 space. According to the authors, there is only one paper (i.e., the paper by Tessitore and Zabczyk in 1998) that provides some sufficient conditions on the correlation measure that guarantees the existence of an invariant measure for the SHE on the whole spatial space. However, I think that other papers deal with invariant measures when the spatial domain is the whole space, e.g., the KPZ equation on the real line and stochastic 2d or 3d Navier-Stokes equations. Gu and Li 2020 consider SHE on R^d and show the convergence to the stationary distribution, but their noise is smooth in space.

Regarding the proof of the main theorem (Theorem 1.4), the authors follow the strategy given in Tessitore and Zabczyk. The essential thing in the proof is to bound uniformly in time the moments of the solution in some weighted norm. For this, the authors used some previous results. More precisely, Reference 10 provided the condition (see (1.10) in the paper) on the correlation measure that guarantees the uniform bound in time of the moment of the solution at each spatial point. This condition (1.10) seems easier to be checked than the one provided by Tessitore and Zabczyk. However, the authors assume one more condition (1.14) on the spectral measure that does not appear in the paper by Tessitore and Zabczyk.

The merit of this paper is that the authors found the conditions on the correlation measure that may be easier to be checked than the condition by Tessitore and Zabczyk. In addition, the conditions for initial functions in this paper cover more general functions including the Dirac delta measure at zero. However, it seems to me that the proofs are fairly standard and they seem to come from the previous results (e.g., the most important thing, which is the condition that produces the uniform bound of the moment of the solution in time, comes from the previous paper), which makes me wonder if this paper meets the very high standards of originality and novelty of TAMS. Furthermore, this article only suggests some sufficient conditions on the correlation measure and the initial function, and I am not so sure if this paper is strong enough to be within 10-12% of all submitted papers to TAMS.